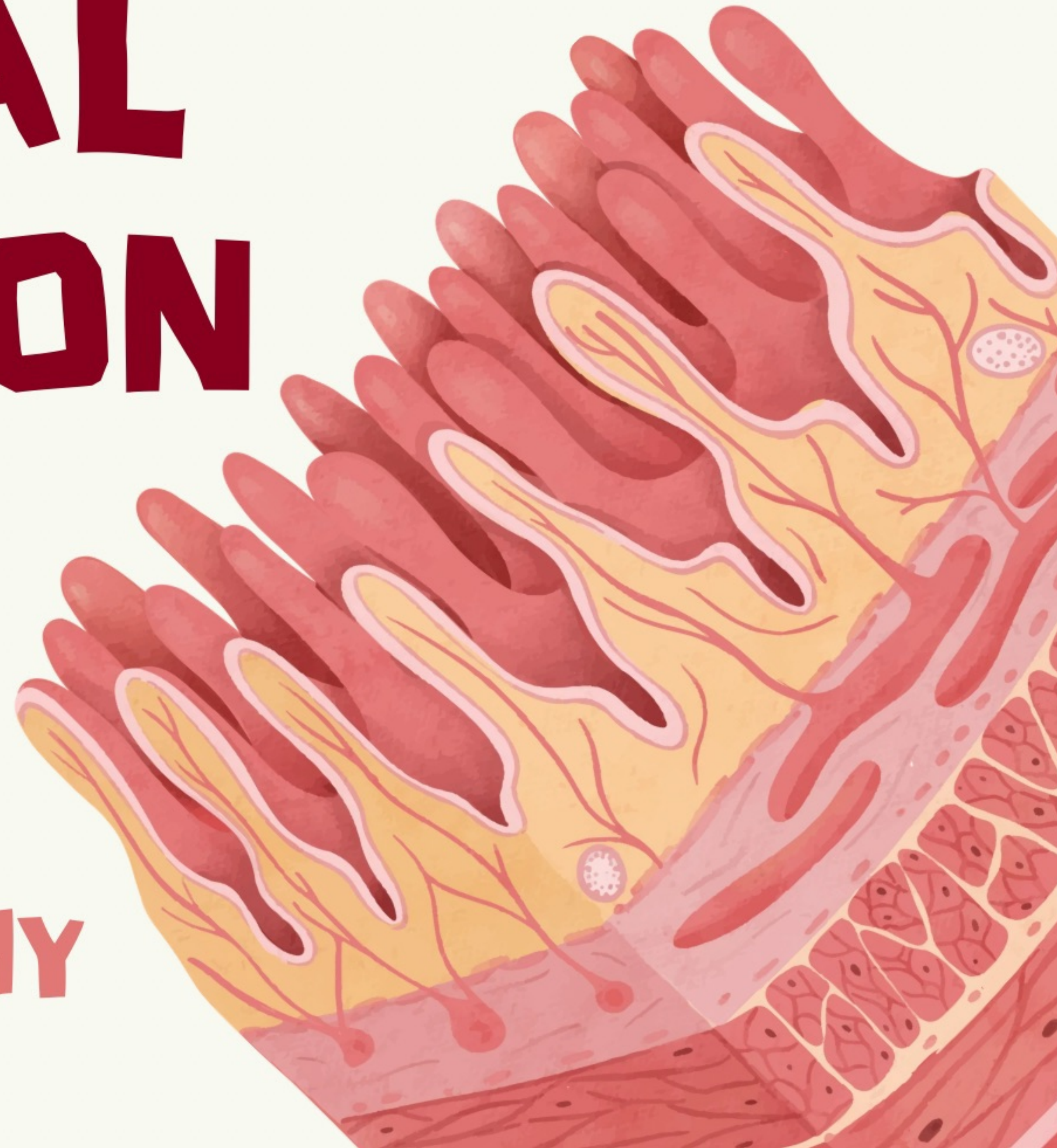


# INTESTINAL ABSORPTION

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## ALL PARTS OF THE GIT CAN ABSORB SUBSTANCES

► **Stomach** → alcohol, water.



► **Large intestine** → water, electrolytes.



► **Small intestine** = chief site of **absorption** due to valvulae conniventes, villi, and microvilli.

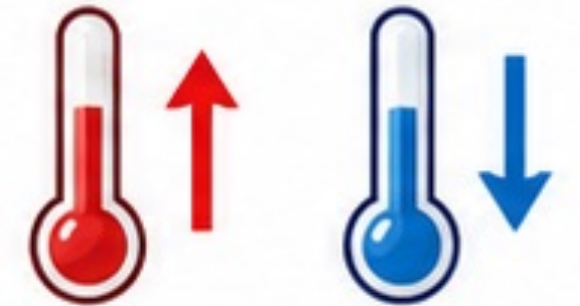


## EVIDENCE THAT INTESTINAL ABSORPTION IS A VITAL (ACTIVE) PROCESS

**1** Intestinal  $O_2$  consumption increases during absorption.



**2** ↑ Temperature → ↑ absorption;  
↓ temperature → ↓ absorption.



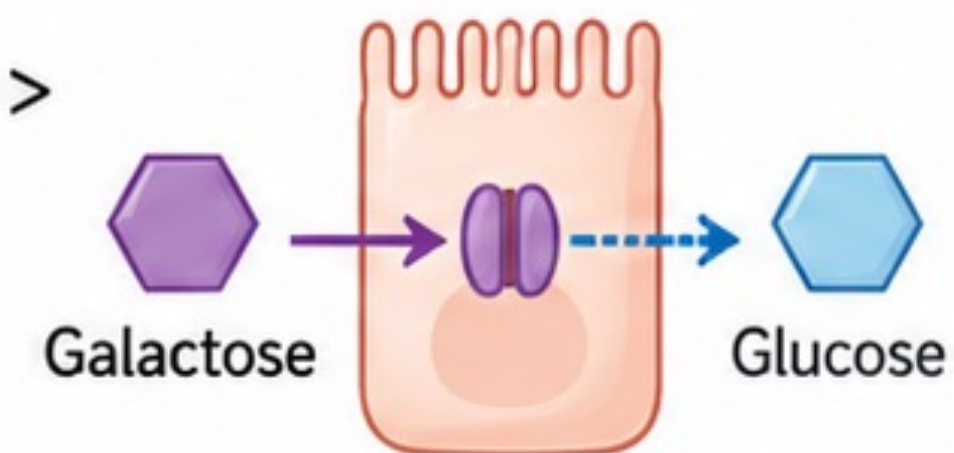
**3** Reduced blood supply markedly decreases absorption.



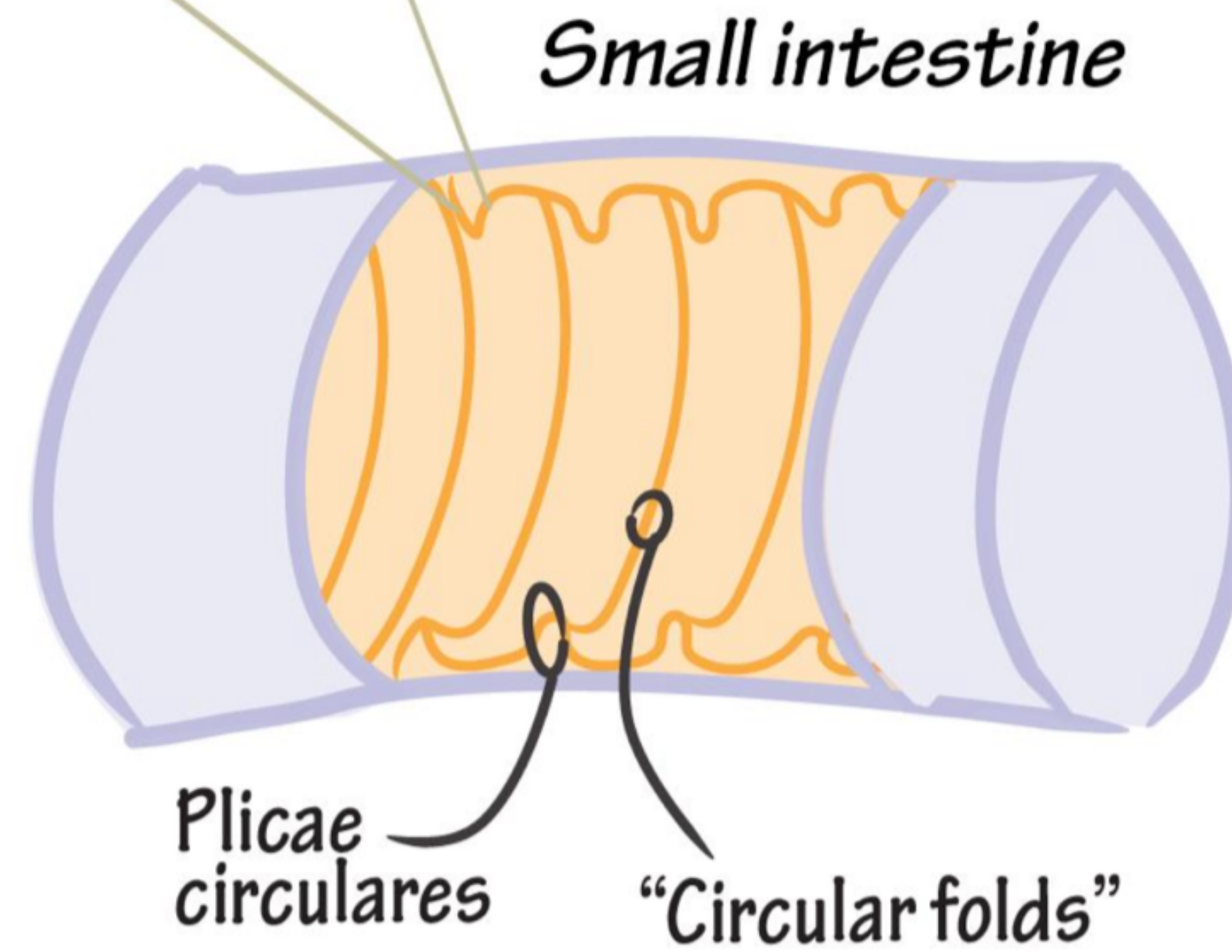
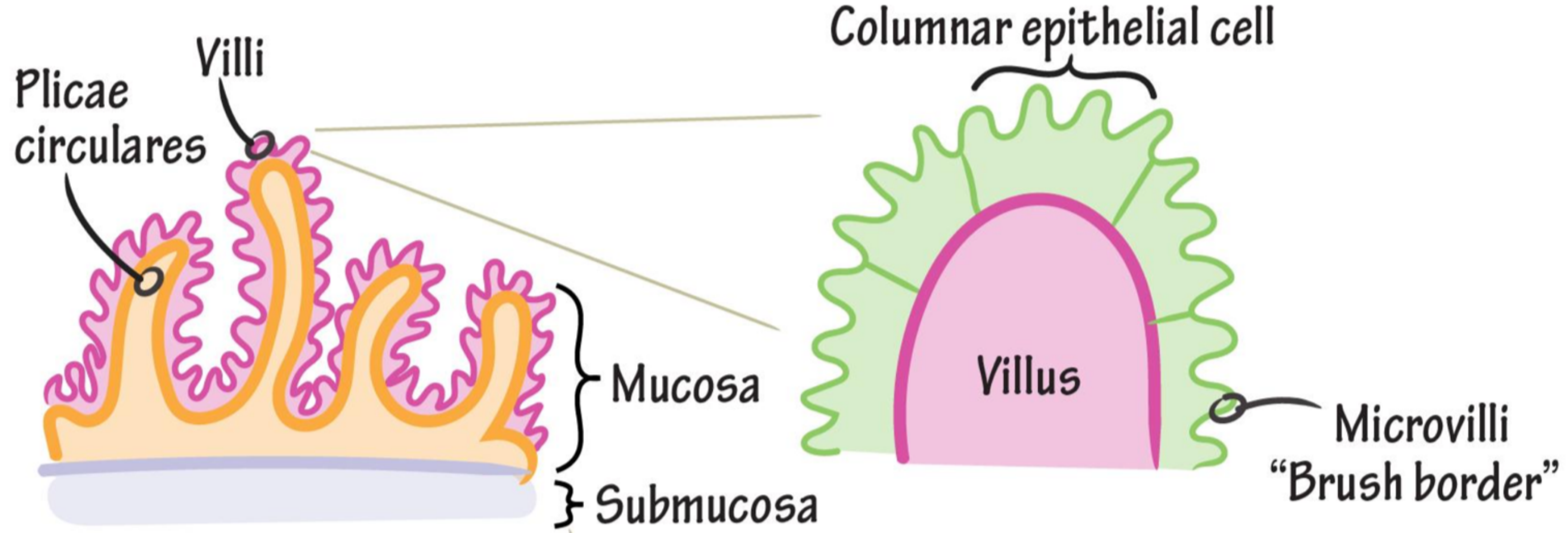
**4** Serum placed in intestine is completely absorbed despite having the same osmolality as plasma.



**5** Selective absorption occurs (galactose > glucose absorption despite equal molecular weight).



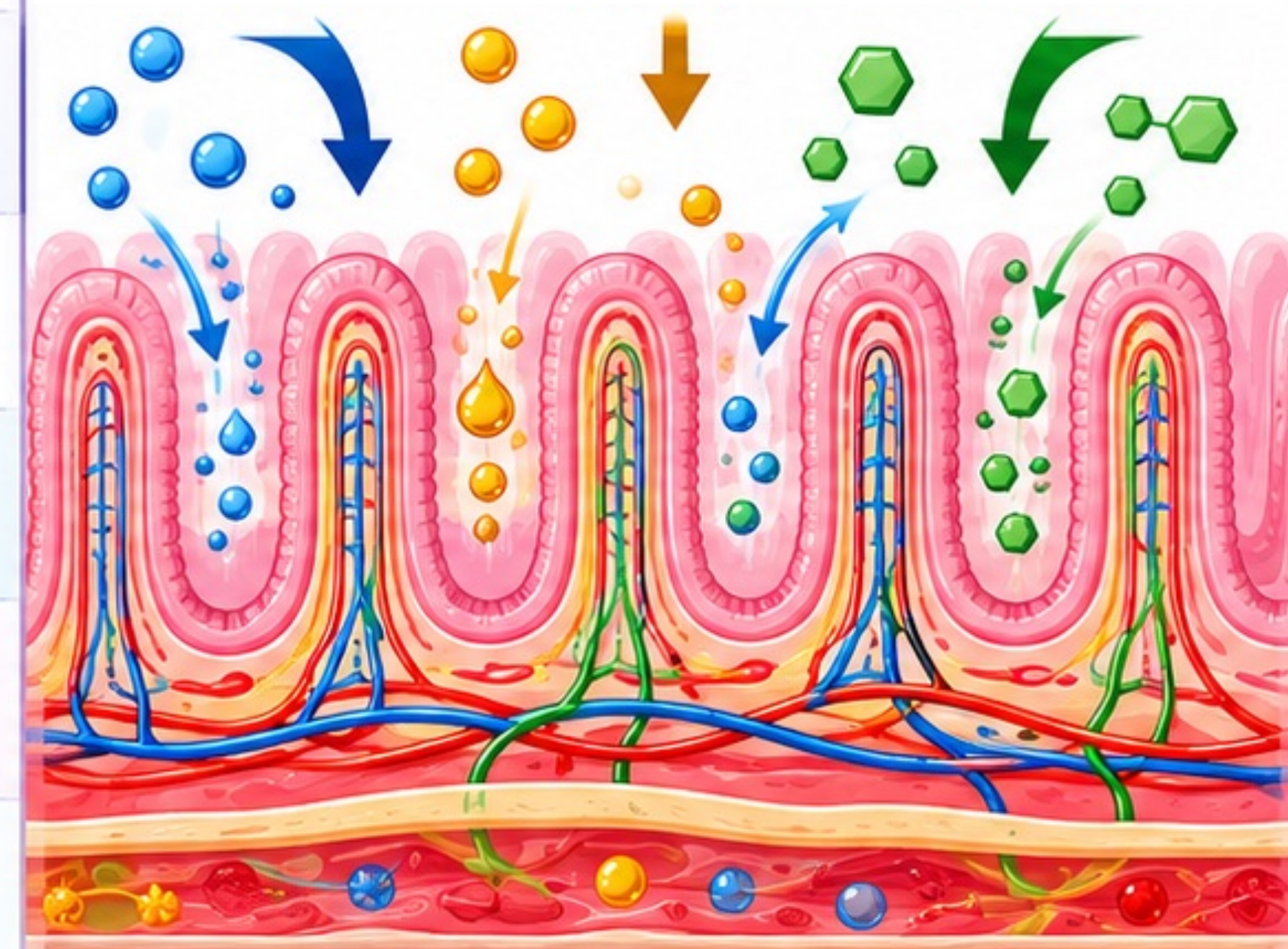






# FACTORS AFFECTING INTESTINAL ABSORPTION

|   |                                                                                     |                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---|-------------------------------------------------------------------------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 |    | <b>Vitality of intestinal mucosa</b>   | <ul style="list-style-type: none"> <li>Requires adequate blood flow, O<sub>2</sub> supply, and vitamins (especially <b>vitamin B</b>).</li> </ul>                                                                                                                                                                                                                                                                                                                                                        |
| 2 |    | <b>State of digestion</b>              | <ul style="list-style-type: none"> <li>Proper digestion is essential for good absorption.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3 |    | <b>Bile salts &amp; lymph flow</b>     | <ul style="list-style-type: none"> <li>Essential for fat absorption.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 4 |    | <b>Duration of contact with mucosa</b> | <ul style="list-style-type: none"> <li>Shortened contact (such as diarrhea) → decreased absorption.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                           |
| 5 |   | <b>Absorptive surface area</b>         | <ul style="list-style-type: none"> <li>Reduction of intestinal surface by &gt;50% markedly decreases absorption.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                              |
| 6 |  | <b>Intestinal mixing movements</b>     | <p>Enhance absorption by:</p> <ul style="list-style-type: none"> <li>Exposing contents to absorptive surface.</li> <li>Improving blood and lymph flow.</li> <li>Increasing intra-intestinal pressure.</li> </ul>                                                                                                                                                                                                                                                                                         |
| 7 |  | <b>Villous movements</b>               | <ul style="list-style-type: none"> <li>Lashing, side-to-side, and pumping (shortening &amp; elongation) movements.</li> <li>Produced by muscularis mucosae.</li> <li>Stimulated mainly by mechanical and chemical irritation.</li> </ul>                                                                                                                                                                                                                                                                 |
| 8 |  | <b>Physicochemical factors</b>         | <ul style="list-style-type: none"> <li><b>Concentration gradient:</b> Passive diffusion increases when luminal concentration &gt; blood concentration.</li> <li><b>Intestinal osmolality:</b> Hyperosmotic solutions ↓ absorption; hyposmotic solutions ↑ absorption.</li> <li><b>Intra-intestinal pressure:</b> Higher pressure → greater absorption.</li> <li><b>Solubility:</b> Greater solubility → greater absorption (acids ↑ Ca<sup>2+</sup> absorption; bile salts ↑ fat absorption).</li> </ul> |



## SMALL INTESTINE RECEIVES



**9000 mL/day**

- 2000 mL ingested fluid
- 7000 mL digestive secretions

## ABSORBED



**8800 mL**

absorbed by small intestine

## EXCRETED



**200 mL**

excreted in stool



# MECHANISM OF INTESTINAL ABSORPTION

## TYPES OF ABSORPTION



- **Active transport:** requires carrier + energy + movement against gradient.



- **Simple (passive) diffusion:** occurs according to concentration and electrical gradients.

## 1. WATER ABSORPTION



- By **osmosis** (simple diffusion) following absorption of electrolytes and nutrients.



- **Stomach:** only a small amount of water is absorbed.



- **Small & large intestines:** water moves in both directions according to osmotic gradients until intestinal contents become isotonic with plasma.



- **Duodenal** contents may be hypertonic or hypotonic depending on the meal, but become nearly isotonic to plasma in the **jejunum**.



## 2. SODIUM (Na<sup>+</sup>) ABSORPTION

About **25–35 g/day** absorbed from the small intestine by **active transport** through:

### A UNIPORT

Active Na<sup>+</sup> pump → blood.

### B SYMPORT

Cotransport of Na<sup>+</sup> with glucose or amino acids via a common carrier.

### C ANTIPORT

Na<sup>+</sup> absorbed in exchange for H<sup>+</sup>.

### MECHANISM

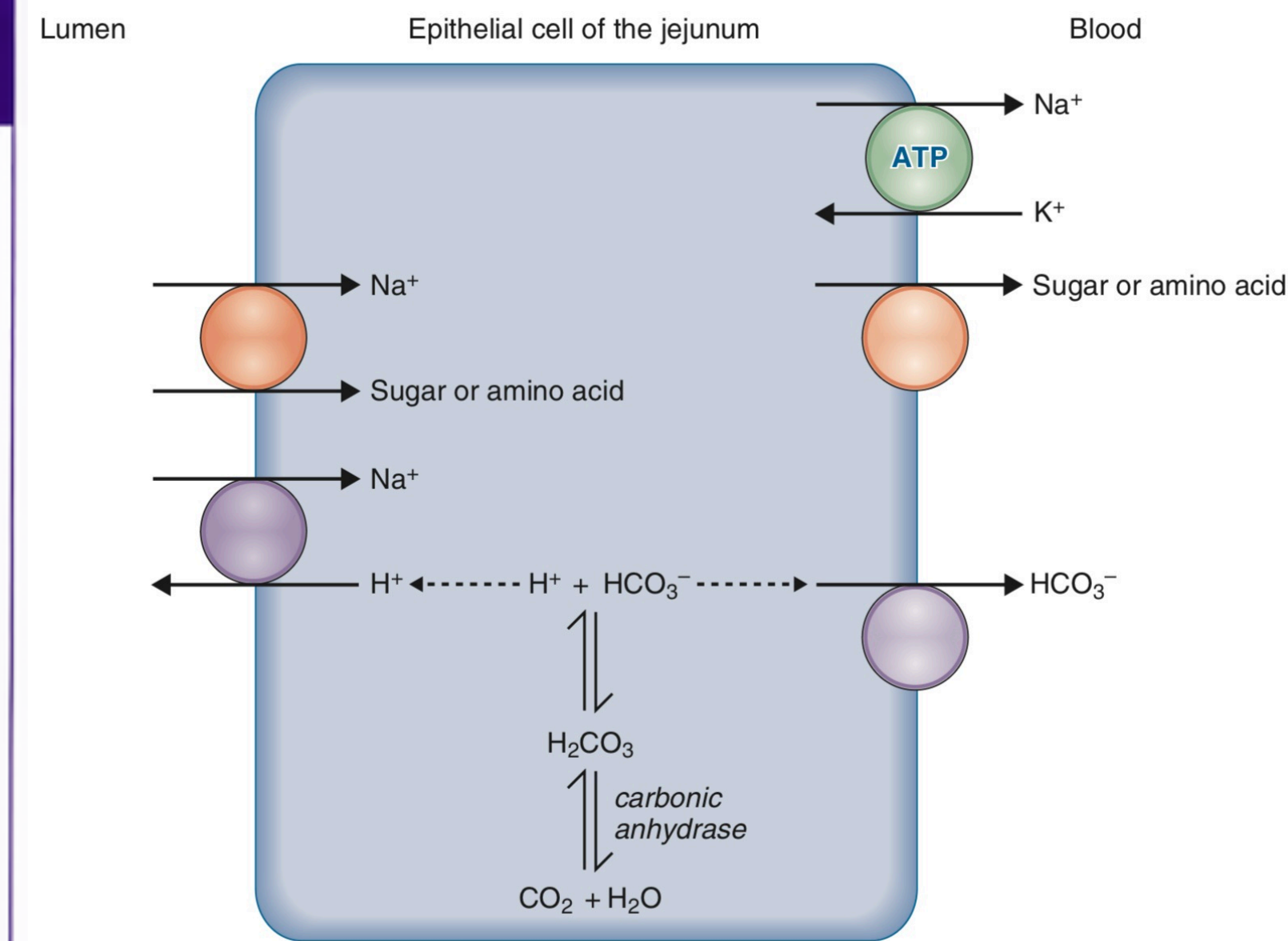
- Na<sup>+</sup> is actively transported to the interstitial fluid in exchange for K<sup>+</sup> at the basolateral membrane → ↓ intracellular Na<sup>+</sup> concentration → Na<sup>+</sup> enters from intestinal lumen through the brush border.

### CLINICAL IMPORTANCE

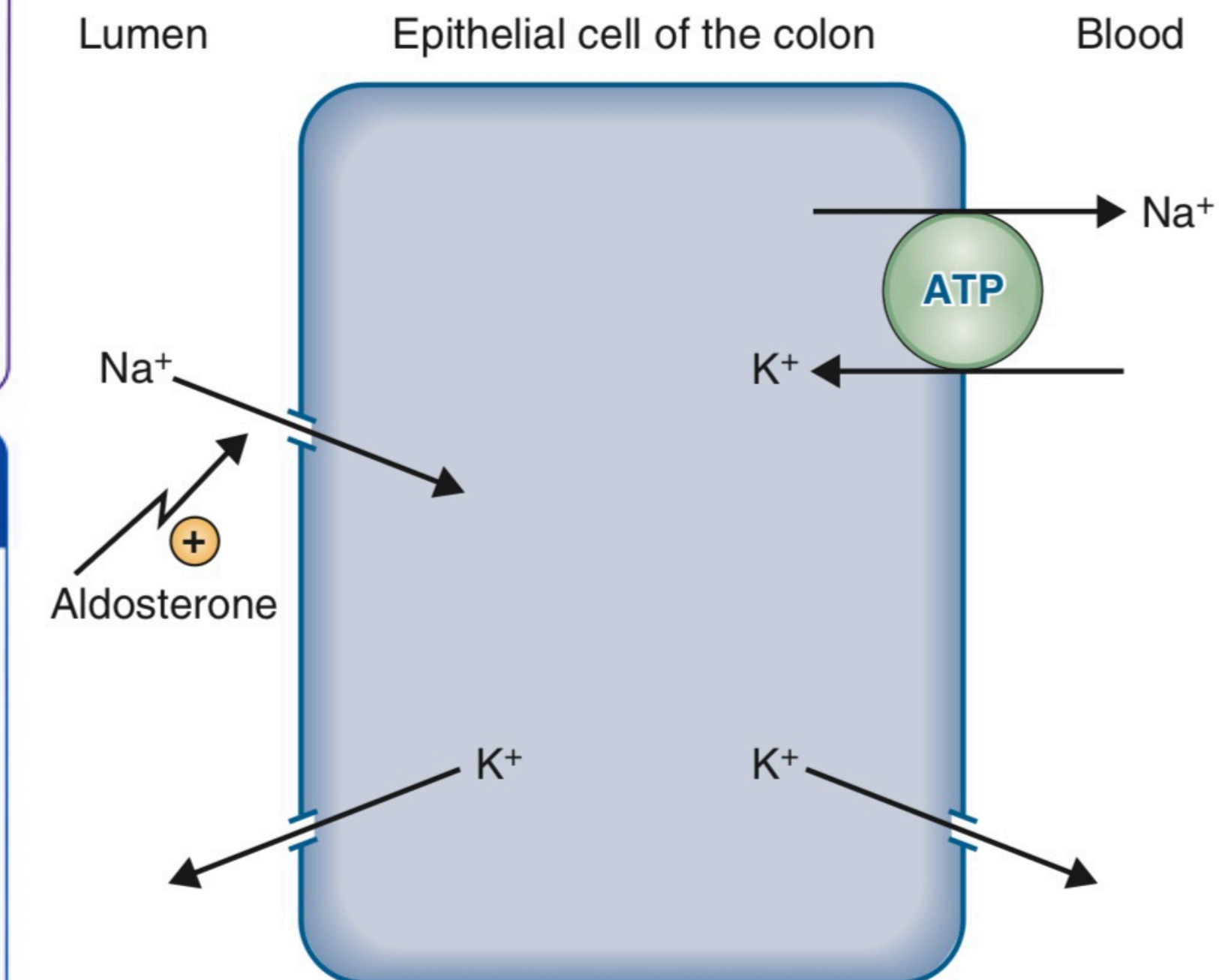
- Na<sup>+</sup> and glucose share the same carrier → facilitate each other's absorption.
- Basis of oral rehydration therapy (ORS) in severe diarrhea: fluids containing NaCl + glucose remain effective because the Na<sup>+</sup>–glucose symporter is intact.

## 3. POTASSIUM (K<sup>+</sup>)

- Some K<sup>+</sup> is secreted with mucus.
- Most K<sup>+</sup> moves passively down its electrochemical gradient into the intestinal lumen, especially in the colon.
- Chronic diarrhea causing loss of ileal and colonic fluids may lead to severe hypokalemia.
- Aldosterone stimulates Na<sup>+</sup> absorption and K<sup>+</sup> secretion via the Na<sup>+</sup>–K<sup>+</sup> pump at the basolateral border of intestinal mucosal cells.



### ELECTROLYTE TRANSPORT IN THE COLON

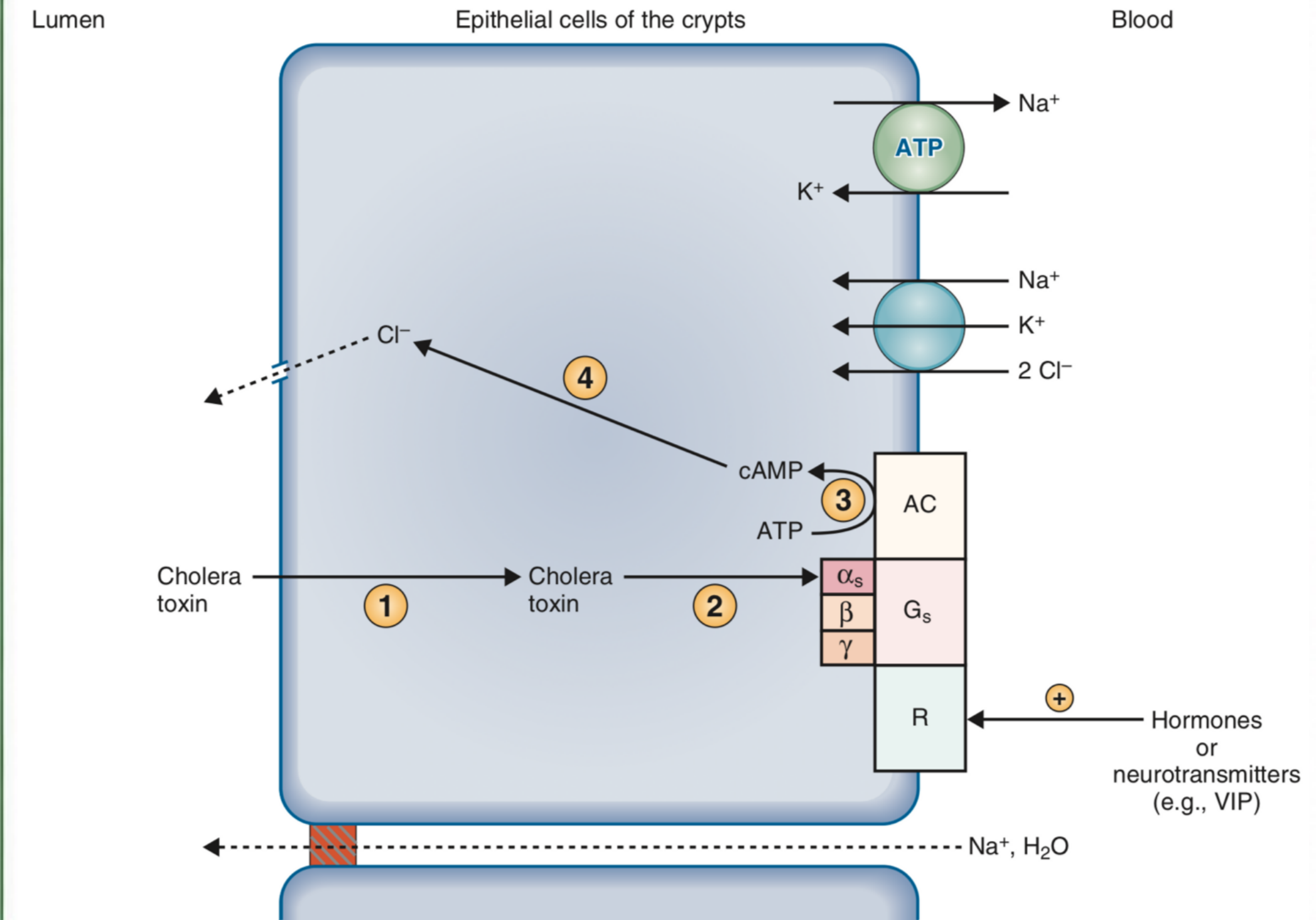




## 4. CHLORIDE (Cl<sup>-</sup>) ABSORPTION

- Absorbed in the small & large intestines mainly by **passive diffusion** following Na<sup>+</sup> absorption (electrical gradient).
- Cl<sup>-</sup> enters enterocytes from interstitial fluid via Na<sup>+</sup>-K<sup>+</sup>-Cl<sup>-</sup> cotransporters (basolateral border) and is **secreted** into the lumen through **cAMP-activated Cl<sup>-</sup> channels**.
- Cholera toxin** activates adenylyl cyclase → ↑ intracellular cAMP → ↑ Cl<sup>-</sup> secretion + ↓ Na<sup>+</sup> absorption.
- Accumulation of NaCl in the lumen causes osmotic water retention → **severe diarrhea**.
- Treatment:** large amounts of **NaCl** + **K<sup>+</sup>** + **glucose** solutions.

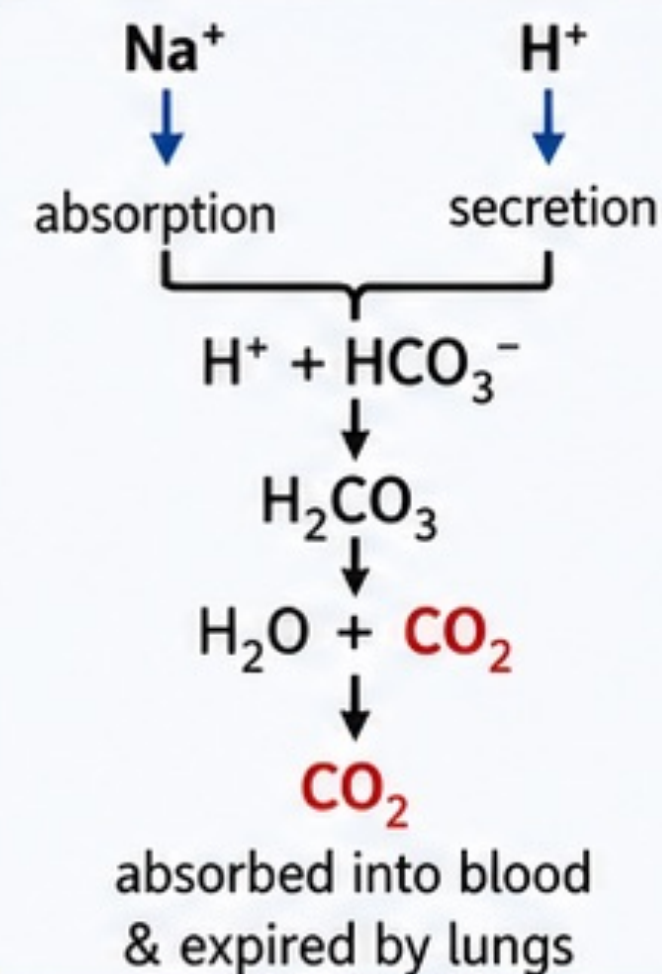
## Cl<sup>-</sup> SECRETION IN THE INTESTINE



## 5. BICARBONATE (HCO<sub>3</sub><sup>-</sup>) ABSORPTION

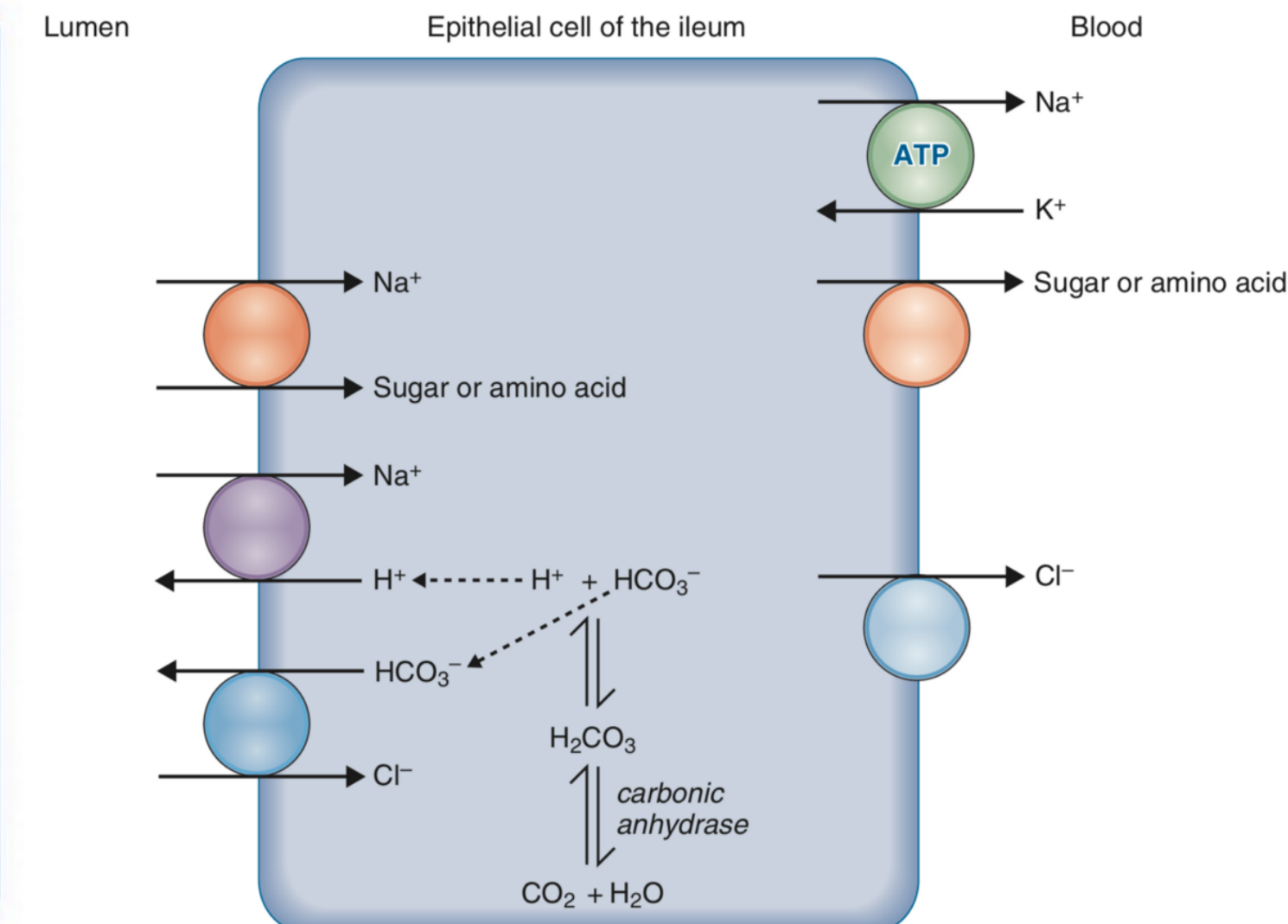
### Duodenum & Jejunum

- Indirect active absorption:**
  - Na<sup>+</sup> absorption → H<sup>+</sup> secretion.
  - H<sup>+</sup> + HCO<sub>3</sub><sup>-</sup> → H<sub>2</sub>CO<sub>3</sub> → H<sub>2</sub>O + CO<sub>2</sub>.
  - CO<sub>2</sub> is absorbed into blood and expired by lungs.

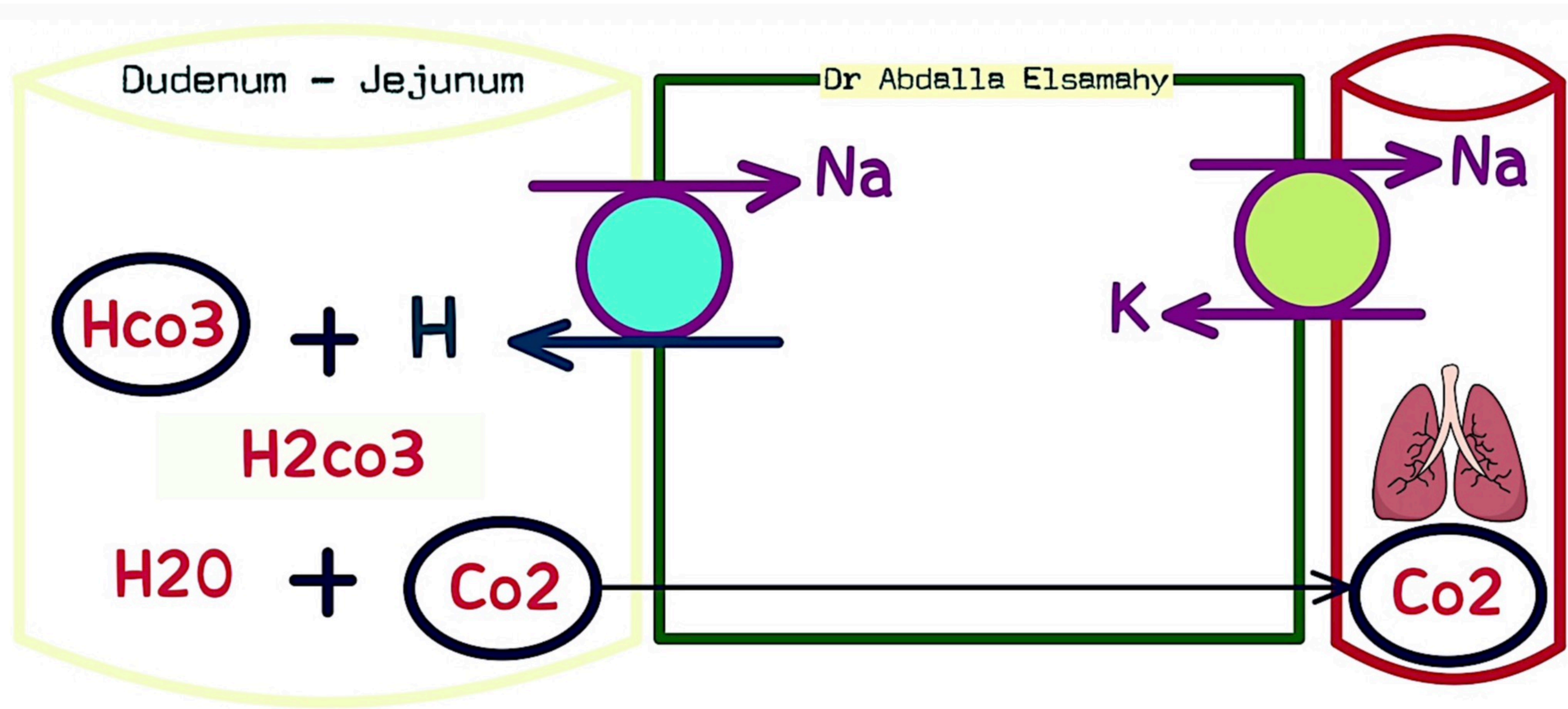


### Ileum & Large Intestine

- HCO<sub>3</sub><sup>-</sup> **secretion** in exchange for Cl<sup>-</sup> absorption.









## 6. CALCIUM (Ca<sup>2+</sup>) ABSORPTION

- 1** Facilitated diffusion at luminal border.
- 2** Active transport at basolateral border.
- 3** Controlled by **Parathormone (PTH)** and **Vitamin D<sub>3</sub>**.

## 7. IRON ABSORPTION



- **Active absorption** in the duodenum.



- Stimulated by **erythropoietin**.

## 8. CARBOHYDRATE ABSORPTION (LUMINAL BORDER)

### Glucose



- Active Na<sup>+</sup>–dependent transport via common Na<sup>+</sup>–glucose carrier.
- Inhibition of Na<sup>+</sup> absorption (by **glycosides**) → ↓ glucose absorption.

### Galactose



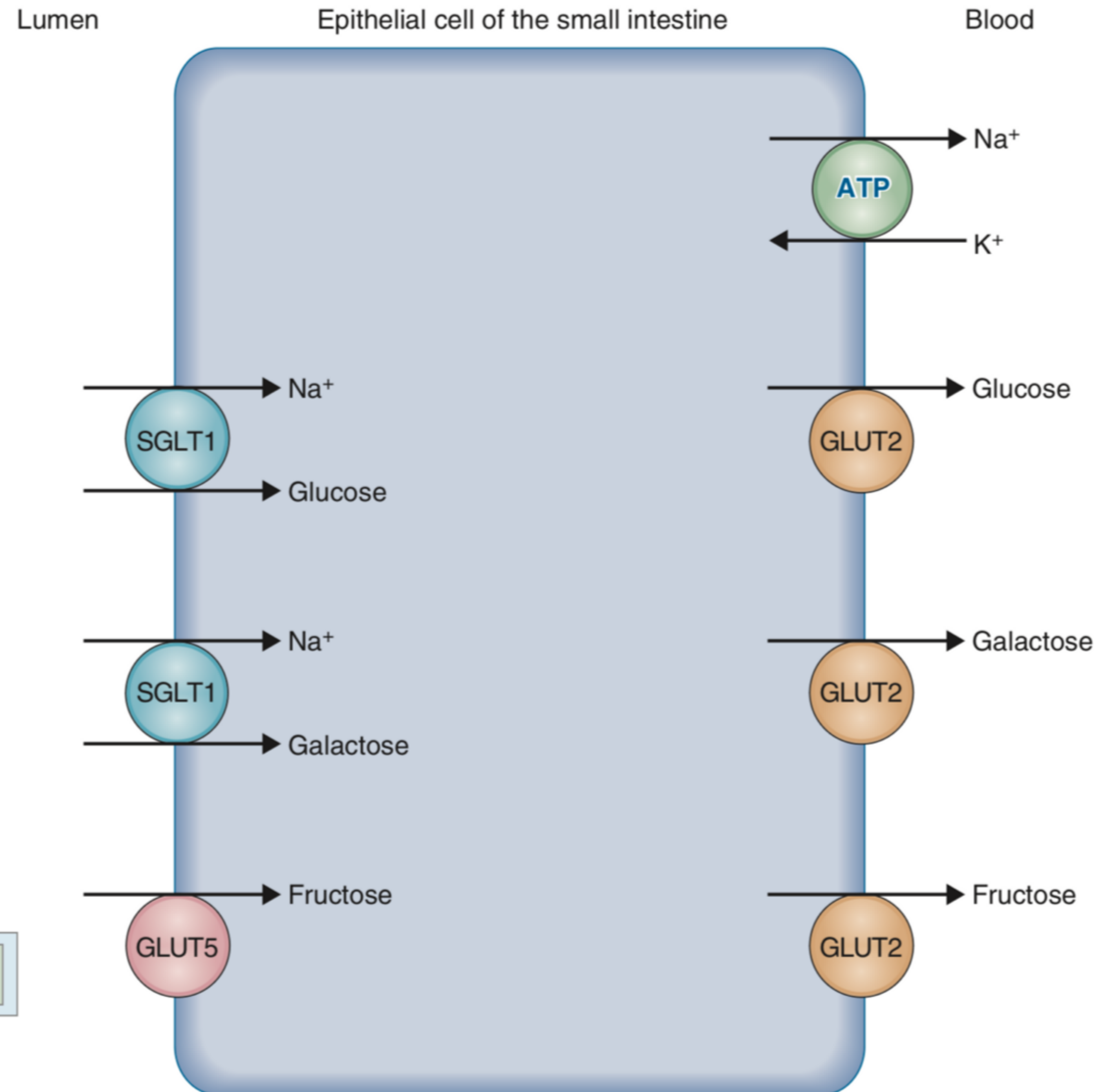
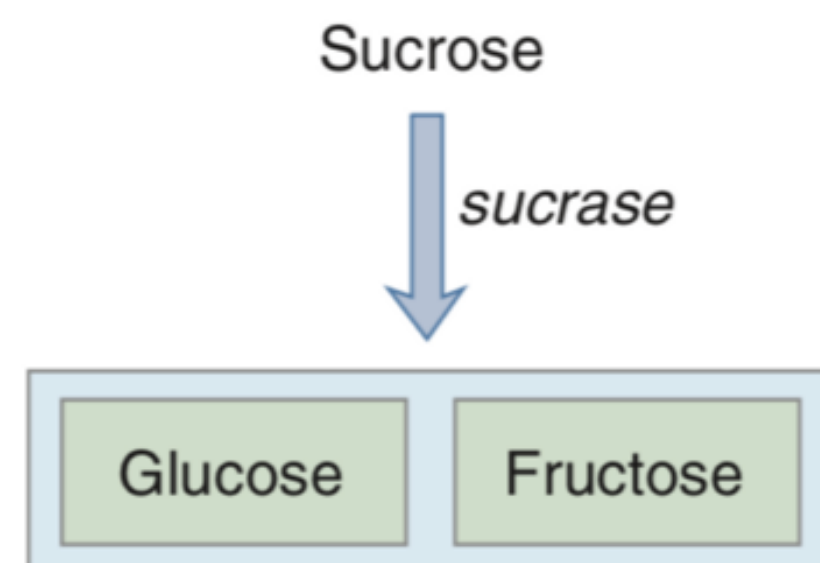
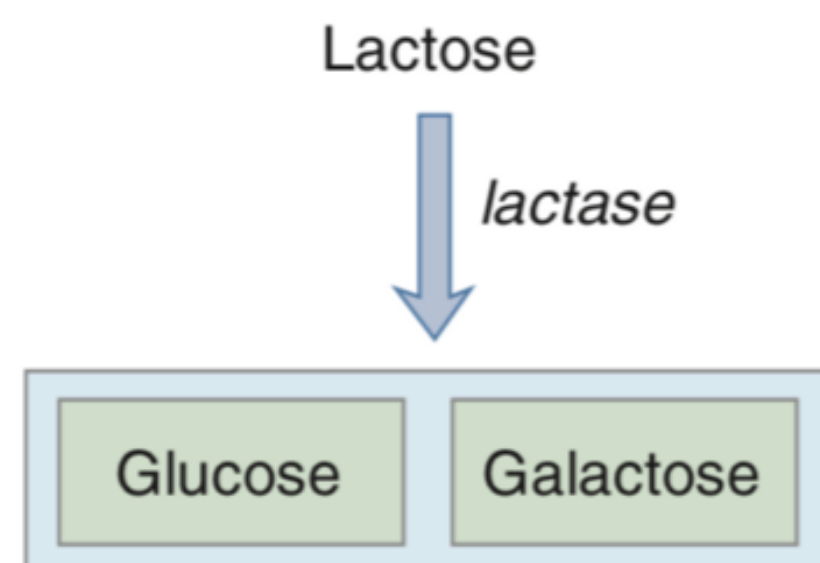
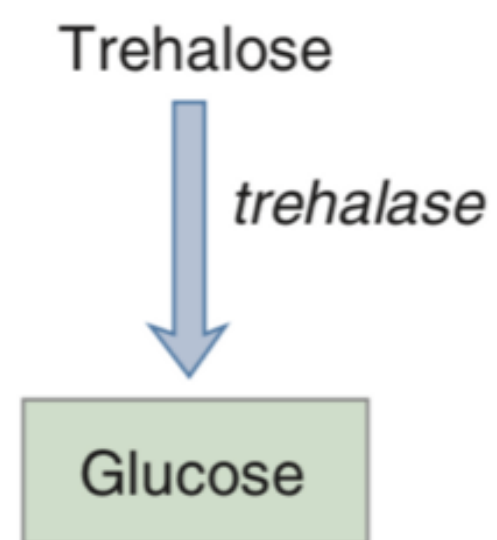
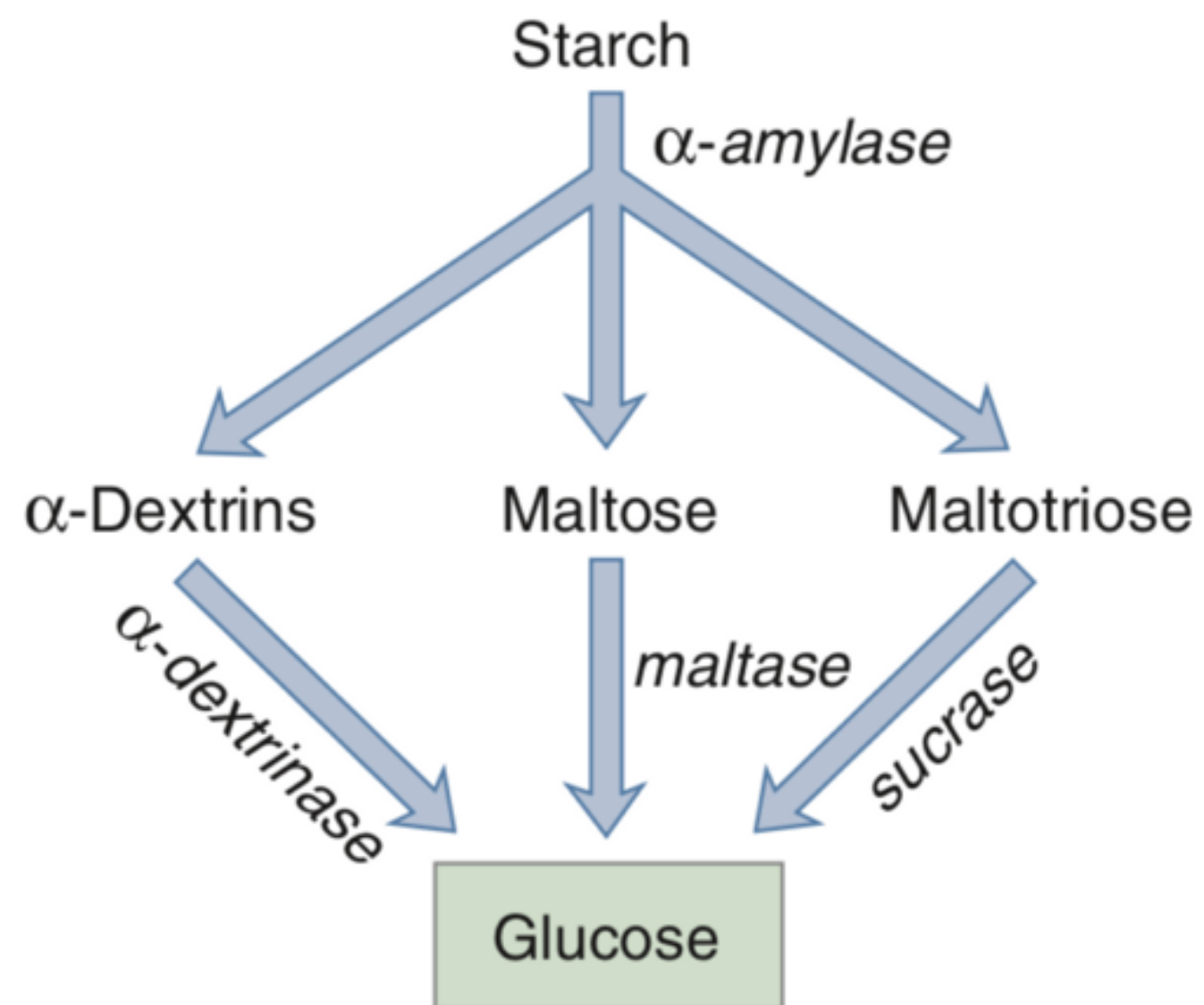
- Same mechanism as glucose (Na<sup>+</sup>–dependent active transport).

### Fructose



- Facilitated diffusion (passive).







# 8

## PROTEIN ABSORPTION

### Amino Acids

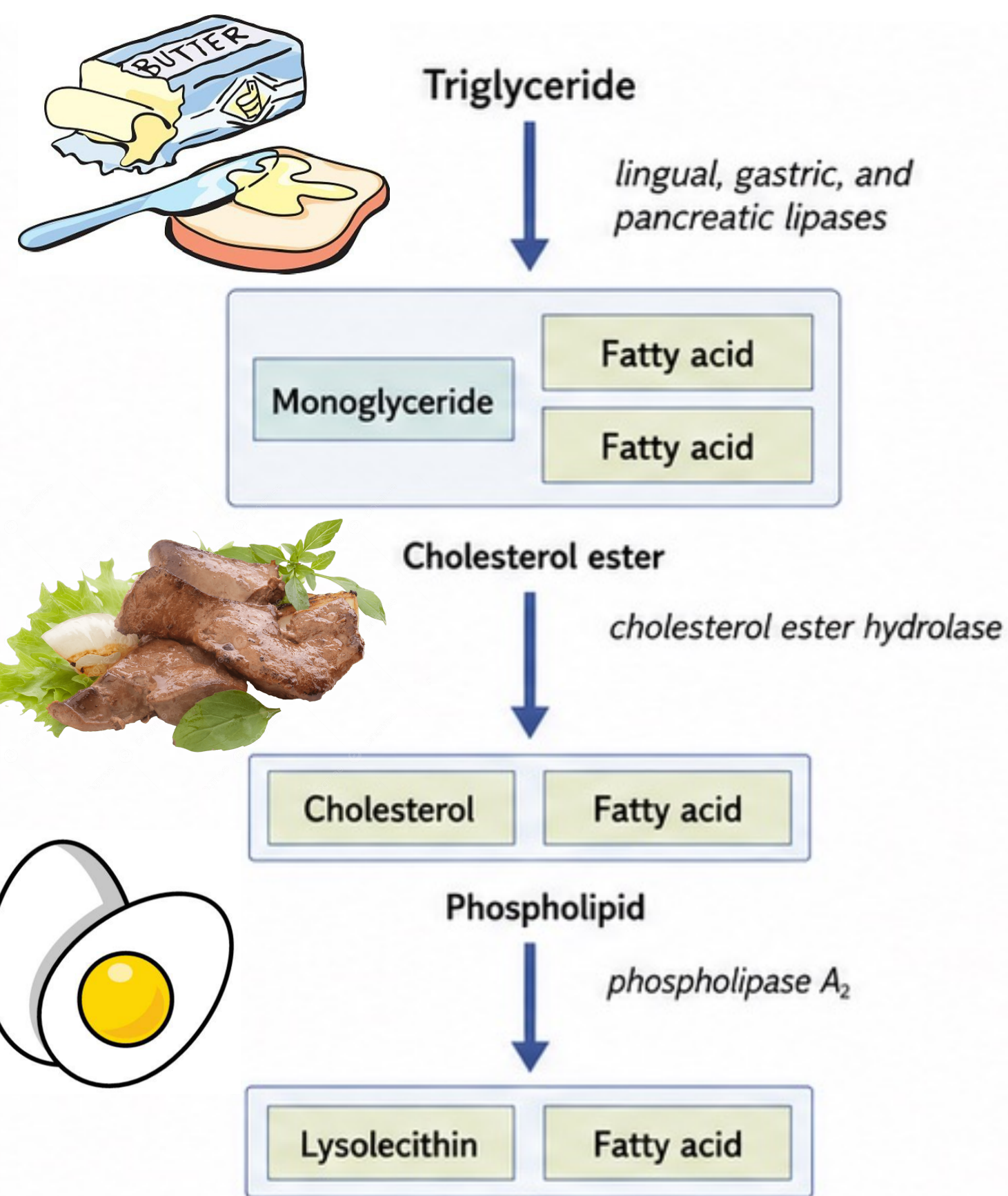


- **L-amino acids** absorbed by  $\text{Na}^+$ -dependent active transport (similar to glucose).
- Uses 4 carrier types for:
  - Neutral amino acids
  - Basic amino acids
  - Acidic amino acids
- Absorbed mainly in the **jejunum**.

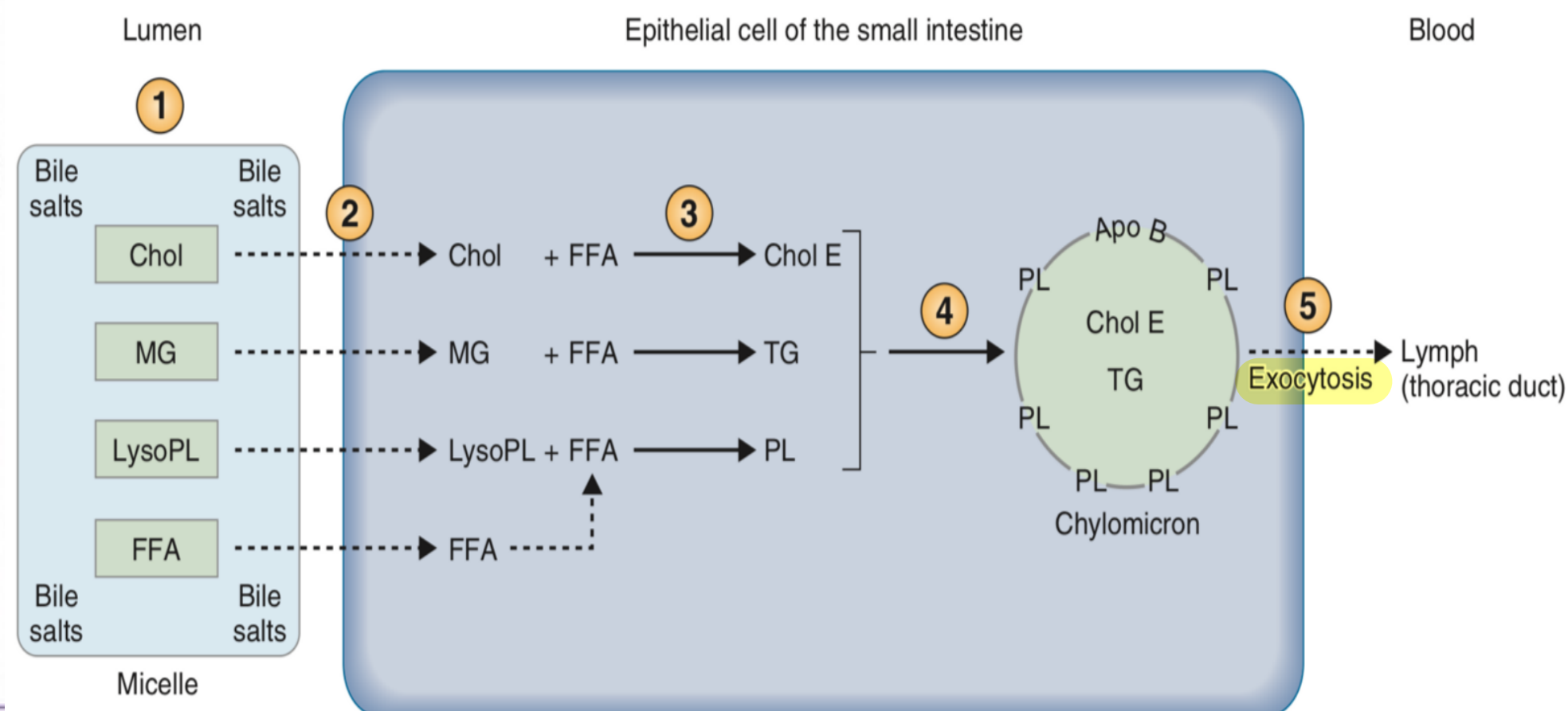
### Intact Proteins



- Small amounts absorbed intact by **pinocytosis** (endocytosis).



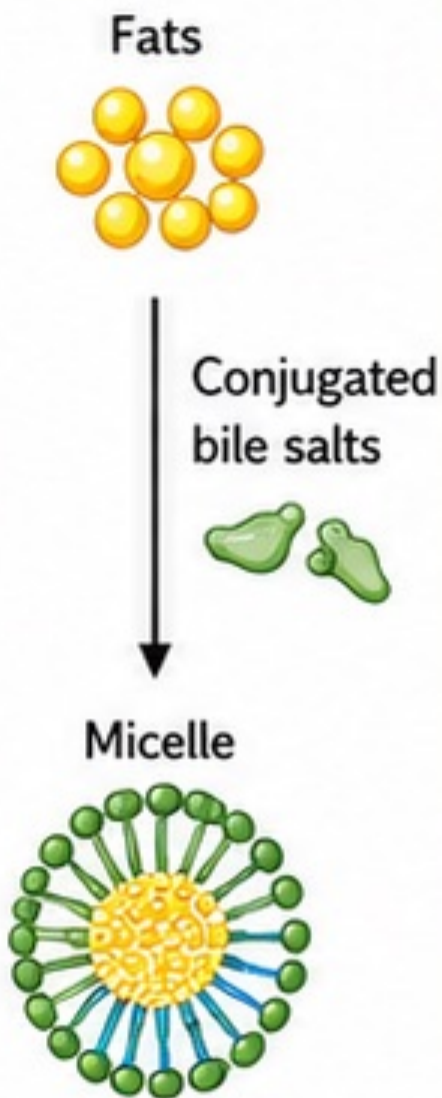
### ABSORPTION OF LIPIDS



# 9

## LIPID ABSORPTION

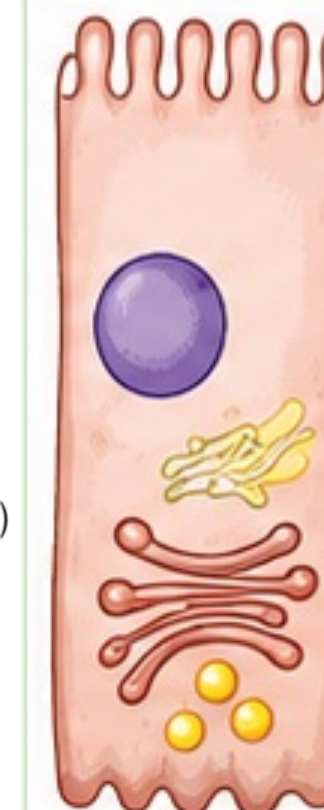
### In Intestinal Lumen



- Conjugated bile salts emulsify fats.
- Lipids form micelles containing:
  - Fatty acids
  - Monoglycerides
  - Cholesterol
  - Fat-soluble vitamins

Micelles are surrounded by a bile salt shell and enter enterocytes by **simple diffusion**.

### Inside Enterocytes



- Short-chain fatty acids → directly to **portal blood**.
- Long-chain fatty acids → re-esterified to **triglycerides**.
- Some cholesterol → **cholesterol esters**.
- Triglycerides + cholesterol esters combine with proteins, cholesterol, and phospholipids in the **Golgi complex** → **chylomicrons**.
- Chylomicrons leave by **exocytosis** and enter **lymphatic vessels**.





# 10

## VITAMIN ABSORPTION

### WATER-SOLUBLE VITAMINS



- Absorbed mainly from the **jejunum** by **simple diffusion**.
- Vitamin **B<sub>12</sub>** requires **intrinsic factor** for absorption.

### FAT-SOLUBLE VITAMINS (A, D, E, K)



- Absorbed by **simple diffusion**.
- Depend on **normal fat digestion** and absorption.

## MALABSORPTION SYNDROME

### CAUSES



- Removal of **>50%** of the intestine.



- **Maldigestion** due to:
  - ↓ Pancreatic secretion → inadequate lipolysis.
  - Obstructive jaundice → ↓ fat and vitamin digestion/absorption.



### CONSEQUENCES



- ↓ Amino acid absorption → body wasting + **edema**.



- ↓ Fat absorption → ↓ absorption of **fat-soluble vitamins (A, D, E, K)**.



- Steatorrhea (fatty stool) and **bleeding tendency**.